

Problem

If the current flowing through an element is given by

$$i(t) = \begin{cases} 3t \text{ A}, & 0 \leq t < 6 \text{ s} \\ 18 \text{ A}, & 6 \leq t < 10 \text{ s} \\ -12 \text{ A}, & 10 \leq t < 15 \text{ s} \\ 0, & t \geq 15 \text{ s} \end{cases}$$

Plot the charge stored in the element over $0 < t < 20 \text{ s}$.

Step-by-step solution

Step 1 of 6

Consider the current through an element is defined as,

$$i(t) = \begin{cases} 3t \text{ A}, & 0 \leq t < 6 \text{ s} \\ 18 \text{ A}, & 6 \leq t < 10 \text{ s} \\ -12 \text{ A}, & 10 \leq t < 15 \text{ s} \\ 0, & t \geq 15 \text{ s} \end{cases}$$

Consider the initial charge is $q(0) = 0$.

Determine the energy stored in element.

Consider the equation for the energy.

$$q(t) = \int_0^t i(t) dt + q(0) \dots\dots (1)$$

Step 2 of 6

Determine the energy for $0 \leq t < 6 \text{ s}$.

For $0 \leq t < 6 \text{ s}$, $i(t) = 3t$.

$$\begin{aligned} q(t) &= \int_0^t 3t dt + 0 \\ &= 3 \left[\frac{t^2}{2} \right]_0^t \\ &= 3 \left(\frac{t^2}{2} - 0 \right) \end{aligned}$$

Determine the energy at $t = 6 \text{ s}$.

$$q(t) = 3 \left(\frac{t^2}{2} - 0 \right)$$

Replace $t = 6 \text{ s}$ in equation.

$$\begin{aligned} q(6) &= 3 \left(\frac{(6)^2}{2} - 0 \right) \\ &= 3 \left(\frac{36}{2} \right) \\ &= 54 \text{ C} \end{aligned}$$

Step 3 of 6

Determine the energy for $6 \leq t < 10 \text{ s}$.

For $6 \leq t < 10 \text{ s}$, $i(t) = 18$.

$$\begin{aligned} q(t) &= \int_6^t 18 \, dt + q(6\text{s}) \\ &= [18t]_6^t + 54 \\ &= 18t - 108 + 54 \\ &= 18t - 54 \end{aligned}$$

$$q(t) = 18t - 54$$

Determine the energy at $t = 10 \text{ s}$.

$$q(t) = 18t - 54$$

Replace $t = 10 \text{ s}$ in equation.

$$\begin{aligned} q(t) &= 18(10) - 54 \\ &= 126 \text{ C} \end{aligned}$$

Step 4 of 6

Determine the energy for $10 \leq t < 15 \text{ s}$.

For $10 \leq t < 15 \text{ s}$, $i(t) = -12$.

$$\begin{aligned} q(t) &= \int_{10}^t -12 \, dt + q(10\text{s}) \\ &= [-12t]_{10}^t + 126 \\ &= -12t + 120 + 126 \end{aligned}$$

$$q(t) = (-12t + 246) \text{ C}$$

Determine the energy at $t = 15 \text{ s}$.

$$q(t) = (-12t + 246) \text{ C}$$

$$\begin{aligned} q(15 \text{ s}) &= -12(15) + 246 \\ &= -180 + 246 \\ &= 66 \text{ C} \end{aligned}$$

Step 5 of 6

Determine the energy for $t \geq 15 \text{ s}$ $10 \leq t < 15 \text{ s}$.

For $t \geq 15 \text{ s}$ $10 \leq t < 15 \text{ s}$, $i(t) = 0$.

$$\begin{aligned} q(t) &= \int_{15}^t 0 \, dt + q(15 \text{ s}) \\ &= 0 + 66 \end{aligned}$$

$$q(t) = 66 \text{ C}$$

Therefore, the charge stored in the element is

$$q(t) = \begin{cases} \frac{3t^2}{2} \text{ C} & \text{for } 0 \leq t < 6 \text{ s} \\ (18t - 54) \text{ C} & \text{for } 6 \leq t < 10 \text{ s} \\ (-12t + 246) \text{ C} & \text{for } 10 \leq t < 15 \text{ s} \\ 66 \text{ C} & \text{for } t \geq 15 \text{ s} \end{cases}$$

Consider the following plot for charge stored in the element.

